

DATASET INSPECTION REPORT

Project Title: Telco Customer Churn Analysis

1. Introduction

This report presents an inspection of the Telco Customer Churn dataset before analysis. The objective is to assess the dataset's structure, quality, completeness, and suitability for analysis while identifying any data quality issues that require cleaning or transformation.

2. Dataset Overview

Attribute	Description
Dataset Name	Telco Customer Churn Dataset
Number of Records	7,043
Number of Columns	21
File Format	CSV
Tool Used	Microsoft Excel and Power BI
Unit of Analysis	One record represents one customer

3. Dataset Features

The dataset contains information about customer demographics, subscription details, account information, and churn status.

Some key variables include:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service

- Internet Service
 - Online Security
 - Online Backup
 - Device Protection
 - Tech Support
 - Streaming TV
 - Streaming Movies
 - Contract
 - Paperless Billing
 - Payment Method
 - Monthly Charges
 - Total Charges
 - Churn
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4. Data Types

The dataset contains different data types, including:

Data Type	Examples
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Text (Categorical)	Gender, Contract, Payment Method, Internet Service
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Numerical	Tenure, Monthly Charges, Total Charges
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Binary	Senior Citizen, Churn
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5. Data Quality Assessment

The dataset was inspected for common data quality issues before analysis.

The following checks were performed:

- Missing values
- Duplicate records
- Incorrect data types
- Blank records

- Data consistency
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6. Data Cleaning Performed

The following data preparation steps were carried out:

- Imported the dataset into Power BI(Power Query)
 - Reviewed each column for missing or inconsistent values.
 - Corrected data types where required.
 - Removed unnecessary records where applicable.
 - Verified numerical fields for accurate calculations.
 - Created calculated measures and KPIs for dashboard analysis.
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7. Dataset Suitability

Following data inspection and cleaning, the dataset was suitable for analysis and dashboard development.

The cleaned dataset was used to generate insights on:

- Customer demographics
 - Customer churn
 - Customer retention
 - Contract analysis
 - Payment methods
 - Internet services
 - Value-added services
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8. Conclusion

The Telco Customer Churn dataset was found to be suitable for business analysis after inspection and cleaning. Data quality checks ensured the reliability of the analysis, enabling meaningful insights into customer behavior and the factors influencing churn.